8. The perimeter of a certain sector of a circle is equal to the length of the arc of a semi-circle having the same radius. Express the angle of the sector in degrees, minutes and seconds.
9. An airplane propeller rotates $1000$ times per minute. Find the number of degrees that a point on the edge of the propeller will rotate in $1$ second.
10. A train is moving on a circular track of $1500\ m$ radius at the rate of $66\ km/hr$. What angle will it turn in $20$ seconds?
11. A circular metallic plate of radius $8\ cm$ and thickness $6\ mm$ is melted and molded into a pie (a sector of the circle with thickness) of radius $16\ cm$ and thickness $4\ mm$. Find the angle of the sector.

## 3.4 Trigonometric functions and their properties

### 3.4.1 Trigonometric Functions of any angle in terms of Cartesian coordinates

We have studied the principles of trigonometric ratios in the lower classes using acute angles . But we come across many angles which are not acute. We shall extend the acute angle idea and define trigonometric functions for any angle. The trigonometric ratios to any angle in terms of radian measure are called trigonometric functions.

Let $P(x, y)$ be a point other than the origin on the terminal side of an angle $\theta$ in standard position . Let $OP = r$. Thus, $r = \sqrt{x^2 + y^2}$

The six trigonometric functions of $\theta$ are defined as follows:

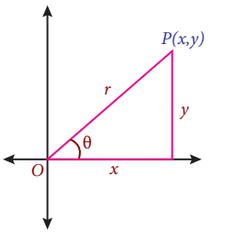


Figure 3.6

$$\sin\theta = \frac{y}{r} \text{ and } \cos\theta = \frac{x}{r}.$$

Using this, we have $\tan\theta = \frac{y}{x}, x \neq 0$; $\cot\theta = \frac{x}{y}, y \neq 0$; $\text{cosec}\,\theta = \frac{r}{y}, y \neq 0$; $\sec\theta = \frac{r}{x}, x \neq 0$.

(i) Since $|x| \leq r, |y| \leq r$, we have $|\sin\theta| \leq 1$ and $|\cos\theta| \leq 1$.
(ii) In the case of acute angle, the above definitions are equivalent to our earlier definitions using right triangle.
(iii) The trigonometric functions have positive or negative values depending on the quadrant in which the point $P(x, y)$ on the terminal side of $\theta$ lies.
(iv) The above definitions of trigonometric functions is independent of the points on the terminal side of the angle. (verify!)

**Trigonometric ratios of Quadrantal angles**

Let us recall that, an angle in its standard position for which the terminal side coincides with one of the axes, is a Quadrantal angle. We shall find the trigonometric ratios for the quadrantal angles.

Consider the unit circle $x^2 + y^2 = 1$. Let $P(x, y)$ be a point on the unit circle where the terminal side of the angle $\theta$ intersects the unit circle.

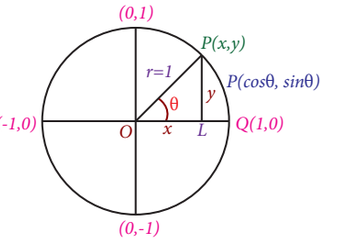


Figure 3.7

Then $\cos\theta = \frac{x}{1} = x$ ($x$-coordinate of $P$) and

$\sin\theta = \frac{y}{1} = y$ ($y$-coordinate of $P$)

Thus, the coordinates of any point $P(x, y)$ on the unit circle is $P(\cos\theta, \sin\theta)$. In this way, the angle measure $\theta$ is associated with a point on the unit circle.

The following table illustrates how trigonometric function values are determined for a Quadrantal angles using the above explanation.